

Pseudocode for Hospital Database Project

This pseudocode outlines the logical steps for designing, creating, and querying the hospital database for assessment 1. It demonstrates the structural planning and the functional logic required to meet the learning outcomes of this module.

Database Creation and Planning

The following pseudocode describes the process of creating and populating the hospital database. It defines the tables, relationships, and data types planned for implementation in MySQL.

START

Step 1: Create the database

CREATE DATABASE hospital_db

CONNECT TO hospital_db

Step 2: Create tables

```
CREATE TABLE Hospital (  
    HospitalID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Beds INT,  
    Accreditation VARCHAR(20)  
)
```

```
CREATE TABLE Doctors (  
    DoctorID INT PRIMARY KEY,  
    Name VARCHAR(100),
```

```
HospitalID INT,  
FOREIGN KEY (HospitalID) REFERENCES Hospital(HospitalID)  
)
```

```
CREATE TABLE Patients (  
    PatientID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    DOB DATE,  
    DoctorID INT  
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID)  
)
```

```
CREATE TABLE Prescriptions (  
    PrescriptionID INT PRIMARY KEY,  
    DoctorID INT,  
    PatientID INT,  
    DrugName VARCHAR(100),  
    DateIssued DATE,  
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID)  
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID)  
)
```

Step 3: Load data into tables

```
READ hospitals.csv  
FOR each record IN hospitals.csv  
    INSERT INTO Hospital (HospitalID, Name, Beds, Accreditation)  
END FOR
```

```
READ doctors.csv  
FOR each record IN doctors.csv
```

```

        INSERT INTO Doctors (DoctorID, Name, HospitalID)
END FOR

READ patients.csv
FOR each record IN patients.csv
    INSERT INTO Patients (PatientID, Name, DOB, DoctorID)
END FOR

READ prescriptions.csv
FOR each record IN prescriptions.csv
    INSERT INTO Prescriptions (PrescriptionID, DoctorID, PatientID,
        DrugName, DateIssued)
END FOR

```

Once completed the code above will have led to the creation of data tables populated with data from the csv files and set up with the correct relationships between data contained in those tables to allow for the effective use of SQL queries.

SQL Query Functionality

The following pseudocode provides outlines each logical step for the SQL queries required for this assignment. It shows how the data will be retrieved and manipulated to meet the six functional tasks.

START

Query 1: Doctors at a particular hospital

```

    INOUT Hospital.Name
    SELECT Doctors.Name
    FROM Doctors
    WHERE Doctors.HospitalID = (

```

```
        SELECT HospitalID FROM Hospital WHERE Name = hospital_name
    )
    OUTPUT list_of_doctors
```

Query 2: Prescriptions for a particular patient

```
    INPUT patient_id
    SELECT DrugName, DateIssued
    FROM Prescriptions
    WHERE PatientID = patient_id
    ORDER BY DateIssued ASC
    OUTPUT list_of_prescriptions
```

Query 3: Prescriptions issued by a particular doctor

```
    INPUT doctor_name
    SELECT DrugName, DateIssued
    FROM Prescriptions
    WHERE DoctorID = (
        SELECT DoctorID FROM Doctors WHERE Name = doctor_name
    )
    OUTPUT doctors_prescriptions
```

Query 4: Add a new patient

```
    INPUT new_patient_details (PatientID, Name, DOB, DoctorID)
    INSERT INTO Patients (PatientID, Name, DOB, DoctorID)
    VALUES (...)
    OUTPUT confirmation_message
```

Query 5: Doctor with the most prescriptions

```
    SELECT DoctorID, COUNT (*) AS PrescriptionCount
    FROM Prescriptions
    GROUP BY DoctorID
```

```
ORDER BY PrescriptionCount DESC  
LIMIT 1  
OUTPUT doctor_with_most_prescriptions
```

Query 6: Doctors at the largest hospital

```
SELECT Name  
FROM Doctors  
WHERE HospitalID = (  
    SELECT HospitalID  
    FROM Hospital  
    ORDER BY Beds DESC  
    LIMIT 1  
)  
OUTPUT doctors_at_largest_hospital
```

END

Together, these pseudocode sections illustrate both the structural design and the functional logic of the hospital database. They provide a clear plan for implementation and testing in MySQL using SQL queries.